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The Completion, For Everyone

The whole theory in plain words: one number from the edge of the universe, and the field theory built to carry it

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Version 1, 2026-08-09.

*This is the plain-language companion to a technical paper called **The Completion** (DOI 10.5281/zenodo.21863521). Nothing in this document is new. Every claim here is the readable version of a claim made precisely there, and every load-bearing number traces back to a named computer script, committed to a public repository, that re-derives the number from scratch and refuses to run — exits with an error — if the number stops being true. Where this document simplifies, the technical paper governs.*

1. The whole story in one page

Galaxies do not spin the way gravity says they should. The stars at their outer edges move far too fast for the visible matter to hold them, and they have been doing so in every galaxy we have ever measured, for fifty years of measurements. Physics has two standard explanations. The first says there is invisible extra matter — *dark matter* — five times more of it than everything we can see, made of a particle nobody has ever detected. The second, called MOND, says there is no extra matter: gravity itself works differently when it gets extremely weak, below a particular threshold

acceleration called a_0 — a threshold about one hundred-billionth of the pull you feel standing on Earth.

MOND's threshold number was always measured from galaxy data and never explained. This program starts from an observation about that number: **it is almost exactly what you get by combining the speed of light with the density of dark energy** — the mysterious ingredient that is pushing the universe's expansion to speed up. Written as a formula: $a_0 = \kappa \cdot c \cdot \sqrt{G \cdot \rho_\Lambda}$, where c is the speed of light, G is Newton's gravitational constant, ρ_Λ is the dark-energy density, and κ is a plain number close to one-half. In other words: **the threshold where galaxies stop obeying Newton appears to be set by the same ingredient that dominates the universe as a whole**. Either that is a coincidence, or it is a clue about what gravity actually is.

A clue is not a theory. A one-line formula cannot tell you how light bends, what the early universe looked like, or whether the idea even makes internal sense. For that you need a *field theory* — the full mathematical machine, the same kind of object Einstein's general relativity is. This document describes that machine: what it is made of, what it gets right, what it costs, how it was stress-tested, and — most importantly — **exactly what is and is not being claimed**, including the things that were claimed earlier in this program, turned out to be wrong, and were publicly withdrawn.

The short version of the result: the machine exists, it is written down in one box, and it has passed every safety and consistency test run on it so far — with two flags still open and several named tests still owed (Sections 11 and 15). It reproduces galaxies, the bending of light, the detailed pattern of the universe's oldest light, and the solar system, and it contains a live — explicitly unconfirmed — candidate mechanism for galaxy clusters, the one place this kind of theory has always struggled. It does this with five dark-sector numbers where the standard picture uses two, so it is not a simpler theory; its claim is different: that dark energy, the dark matter phenomenon, and the galaxy-scale anomaly are **three faces of one field** rather than three separate mysteries. And it makes predictions sharp enough to kill it.

That is the story. The rest of this document earns it.

2. The problem: galaxies spin wrong

Take the solar system. Mercury, close to the Sun, orbits fast; Neptune, far out, orbits slowly. That falloff is Newton's law doing its job: gravity weakens with distance, so distant things orbit lazily. Every planet obeys it to exquisite precision.

Now zoom out to a spiral galaxy — a hundred billion stars in a disk. The same rule should apply: stars near the bright center should orbit fast, stars at the dim outskirts slowly. Astronomers first checked this properly in the 1970s, and the answer has been replicated thousands of times since: **the outer stars do not slow down**. The orbital speed rises from the center and then goes flat, staying flat as far out as anyone can measure — often far beyond the last visible star, traced by cold hydrogen gas that radio telescopes can see. If Newton's law and the visible matter were the whole story, those outer stars would be flung off into space. They are not.

Something is wrong with either the matter inventory or the law.

Option one — more matter. Wrap every galaxy in a huge invisible halo of new particles that feel gravity but do not emit light, and the fast outer orbits are explained: the stars are responding

to mass we cannot see. This is dark matter, and it grew into the backbone of modern cosmology. In the standard model of the universe, called Λ CDM (Λ is the symbol for dark energy, CDM stands for cold dark matter), about 27% of the universe is this cold dark matter, about 68% is dark energy, and everything we have ever seen — stars, planets, gas, people — is the remaining 5%.

Option two — a different law. In 1983 the physicist Mordehai Milgrom noticed something the halo picture has never explained naturally: the anomaly does not switch on at a particular *distance* or a particular *mass*, but at a particular *acceleration*. Wherever gravity’s pull drops below about 10^{-10} meters per second per second — one part in a hundred billion of Earth’s surface gravity — the anomaly appears, in every galaxy, big or small, dense or diffuse. Above that threshold, Newton holds. Below it, the effective pull is stronger than Newton predicts, in a specific mathematical way. Milgrom called the idea MOND (MODified Newtonian Dynamics), and the threshold acceleration a_0 .

Here is why option two refuses to die, forty years on. If you plot, for thousands of measured points in hundreds of galaxies, the gravity you *observe* against the gravity the *visible matter alone* would produce, the points collapse onto a single tight curve — the same curve for every galaxy. This plot is called the **radial acceleration relation**, and it is one of the tightest empirical laws in extragalactic astronomy. In the halo picture there is no obvious reason a messy, history-dependent process — every galaxy with its own accidents of formation, collision, and gas flow — should conspire to put every galaxy on one universal curve keyed to one universal acceleration. In MOND, that curve *is the law*, so its universality is automatic.

And yet MOND, as Milgrom wrote it, has two famous holes. It is a rule for slow-moving matter only — it says nothing about light, or the early universe, or the cosmos as a whole. And it under-predicts the mass of galaxy *clusters*, the largest bound structures in the universe: even with MOND’s boost, clusters come up short by roughly a factor of two, so MOND traditionally needs some dark matter after all, exactly where its slogan says there should be none.

Both camps, then, live with an embarrassment. The dark-matter camp cannot explain why galaxies obey one universal acceleration law. The modified-gravity camp cannot cover clusters or cosmology. This program is an attempt to build the theory that has neither embarrassment — anchored to one specific idea about where a_0 comes from.

3. The coincidence at the center of everything

Milgrom himself noticed, already in 1983, that his measured threshold a_0 is numerically close to quantities built from the universe’s expansion rate. In modern terms, the sharpest version goes through dark energy.

Dark energy is the ingredient, discovered through supernova surveys in 1998, that makes the universe’s expansion accelerate. Its density, ρ_Λ , is measured. Take that density, combine it with Newton’s constant G and the speed of light c — the only combination with the units of an acceleration is $c \cdot \sqrt{G \cdot \rho_\Lambda}$ — and you get a number in the same neighborhood as the measured a_0 . The relation this program is built on is:

$a_0 = \kappa \cdot c \cdot \sqrt{G \cdot \rho_\Lambda}$, with κ a pure number, measured from galaxy data to be 0.551 ± 0.043 — consistent with exactly one-half.

An equivalent way to write the same statement: $a_0 = c_\Lambda / Z$, where H_Λ is the expansion rate the universe is heading toward as dark energy takes over completely, and $Z = 2\sqrt{8\pi/3} \approx 5.79$.

Both forms are the same claim wearing different clothes. Numerically, the canonical value used throughout this program is $a_0 = 9.36 \times 10^{-11} \text{ m/s}^2$. (Readers comparing with the MOND literature will see a_0 quoted there as about 1.2×10^{-10} — a fitted value roughly 25% higher. The two sit well within the spread of published a_0 measurements, and which one the data actually prefer is one of the measurement questions this program works on; Section 13 returns to it.)

Two honesty notes, immediately, because this is where a program like this can start lying to itself.

First: the *form* of the relation is forced; the *number* κ is not. There is a small uniqueness theorem in the technical corpus: if a_0 is built from G , c , and a density, the combination $c \cdot \sqrt{G \cdot \rho}$ is the *only* one dimensionally possible — the arithmetic of matching units has exactly one solution. So the shape of the formula is mathematics. But the pure number κ in front is **measured, not derived**. Years of attempts to derive $\kappa = \frac{1}{2}$ from first principles are documented in this program’s repository, and every one failed or turned out to be circular — a fact the technical papers state in bold type. The measured value 0.551 ± 0.043 is consistent with $\frac{1}{2}$, but it is also consistent with $1/\sqrt{3}$ and other simple candidates. The honest statement is “ κ is measured, and $\frac{1}{2}$ sits comfortably inside the error bar” — nothing stronger. Section 15 returns to this.

Second: a coincidence of numbers is cheap; the question is what follows. Plenty of numerical coincidences in physics turned out to mean nothing. The only way to find out whether this one means something is to take it seriously as a foundation, build the full theory that carries it, and let that theory make predictions that can fail. That is what “the Completion” means: not a new slogan, but the completed machine.

Why might dark energy plausibly set a galaxy-scale threshold at all? Here is the intuition, offered as intuition only. Dark energy gives the universe a *horizon* — a farthest distance from which light can ever reach us, because beyond it, accelerating expansion carries space away too fast. A horizon is not nothing: in modern physics horizons carry temperature and entropy, as Hawking and others showed for black holes. And there is a curious fact about gentle acceleration: the more gently a body accelerates, the larger the region of spacetime that matters to its motion — until, below roughly c times the horizon’s expansion rate, that region grows as large as the horizon itself. It is not absurd that dynamics should change there; the threshold acceleration at which a falling body starts to “feel the edge of the universe” is, up to a pure number, exactly $c \cdot \sqrt{G \cdot \rho_\Lambda}$. The technical corpus contains a derivation in this spirit that produces the precise *shape* of MOND’s transition function — the mathematical bridge describing how gravity crosses over from Newton above a_0 to the modified law below it (it turns out to reproduce, exactly, a formula Milgrom published in 1999) — while *failing* to produce the measured coefficient, predicting one about twelve times too large, which the galaxy data rule out at 15.6 sigma. That split result — the shape derivable, the coefficient stubbornly not — is one of the cleanest honest findings of the whole program, and it is why κ is treated as a measured quantity.

4. Why a formula is not enough: what a field theory is

Everything so far could be written on an index card. Why do the technical papers run to hundreds of pages of derivations and scripts?

Because an index card cannot answer questions like these:

- **How does light bend?** MOND’s rule covers slow-moving matter. Light is not slow. Galaxies and clusters bend light measurably (gravitational lensing), and any theory that gets lensing wrong is dead on arrival. An earlier arm of this very program — a version where the modification lived in the *inertia* of matter rather than in gravity — was killed by exactly this test: it predicted that the mass a galaxy shows through lensing should differ from the mass it shows through orbits by a factor of about six, and observation says the two agree. That arm was excluded at 21 sigma — a statistical death sentence — and the program said so publicly and moved to the arm this document describes. Theories are supposed to be killable. That one was killed.
- **What happened in the early universe?** The cosmic microwave background (the CMB) — the universe’s oldest light, released 380,000 years after the Big Bang — carries a pattern of hot and cold spots whose detailed statistics are measured to a fraction of a percent. The standard model fits that pattern using cold dark matter. Any replacement theory must reproduce it or die.
- **Is the theory even self-consistent?** Field theories can hide fatal diseases — “ghosts,” instabilities, faster-than-light signals. A formula cannot be checked for these; a full theory can and must be. Section 11 is entirely about this.

A *field theory* answers such questions because it specifies everything, everywhere. A field is a quantity with a value at every point of space and time — the temperature map on a weather chart is a field; so is the electromagnetic field carrying this document to your screen. Einstein’s general relativity is a field theory in which the field is the geometry of spacetime itself. To complete this program, one must write down the *action* — the single master expression from which, by a standard mathematical procedure (the same handle Einstein’s theory turns), every equation of motion for every situation follows. Once the action is fixed, there is no more room to make things up: the theory says what it says in every regime, and either survives the data or does not.

The action of this theory fits in one box. Here it is, described in words; the technical paper gives the symbols. Spacetime geometry appears exactly as in Einstein’s theory. Matter — stars, gas, light, you — appears exactly as in Einstein’s theory, and couples to geometry exactly as in Einstein’s theory (this single choice is why light-bending comes out right, as Section 7 explains). To this, add **two new fields and one function**, described next.

5. The machine: three fields, one new function

The theory is called **AeST** — Aether-Scalar-Tensor — a framework published in 2021 by Constantinos Skordis and Tom Złośnik, who proved something remarkable: theirs is the only known relativistic MOND-type theory able to reproduce the cosmic microwave background. Honesty first, as always: **the AeST framework is theirs, not this program’s**. What is this program’s own is the *choice of the free function* inside it — the part Skordis and Złośnik deliberately left open — plus the coefficient that anchors it to dark energy, and the theorems and mechanisms of Sections 8–11. Think of AeST as an engine block that its designers shipped with the fuel map unspecified — the fuel map being the table that tells an engine exactly how much fuel to deliver in every condition. This program’s claim is a particular fuel map, and everything that follows from it.

The three ingredients:

The metric (the “tensor”). Spacetime geometry, exactly as in general relativity. Nothing modified here.

The aether (the vector field, written $\mathbf{A}^\mu$). Despite the antique name, this is a modern, precise object: at every point of spacetime, an arrow of fixed unit length pointing into the future. A field of compass needles for time, defining at each point a natural notion of “at rest.” Einstein’s relativity famously abolished any *built-in* preferred rest frame; here one returns, but as a dynamical field with its own equations — free to tilt and respond — rather than as rigid scaffolding. This is a real conceptual price. Some of the observational tests it raises are passed cleanly — light-bending and the speed of gravitational waves, as Sections 7 and 11 explain; the precision “preferred-frame” tests from the solar system and pulsars have not yet been computed for this theory, as Section 15 discloses.

The khronon (the scalar field, written φ). A single number at every point — from the Greek *chronos*, time, because it behaves like a universal clock-reading field. This field is the workhorse of the theory: its behavior in different environments *is* dark energy, *is* the dark-matter phenomenon, *is* the MOND effect, depending on where you look.

From the aether and the khronon’s rate of change, the theory builds two basic quantities: one (called Q) measuring how fast the khronon ticks along the time direction, and one (called Y) measuring how steeply it varies across space. The entire content of the theory — everything this program claims as its own — is one function $\mathcal{F}(Y, Q)$ telling spacetime how much energy the khronon’s configuration carries. That function is the fuel map. It has three pieces, and each piece is one of the theory’s three jobs:

= (galaxy piece - depends on Y)	← the MOND kernel, Section 6
+ (cosmology piece - depends on Q)	← dark energy and the cosmic dust, Section 8
+ (cluster piece - couples Y and Q)	← the environment response, Section 10

(The technical paper writes the third piece as $A \cdot B(Y/a_0^2) \cdot (Q - Q_0)^2$, where Q_0 is the khronon’s natural resting ticking rate, B is a bell-shaped curve, and A is an amplitude — all of it Section 10’s subject.) Sections 6, 8, and 10 take the three pieces in turn. Section 7 explains why light behaves. Section 9 answers the question every reader of this program eventually asks — *so is there dark matter in this theory or not?* — with the exact, honest answer.

6. Job one: galaxies — the MOND kernel

The first piece of the function depends only on Y — on how steeply the khronon varies across space — and it is built to do exactly one thing: reproduce, in the regime of galaxies, the framework’s own MOND-type law with the dark-energy-anchored a_0 in it.

The specific transition curve the program uses — its *kernel*, this program’s word for the chosen interpolation between the Newtonian and modified regimes, fixed in advance and registered (more on registration in Section 13) — has three properties worth stating in words:

- **Above the threshold, Newton — overwhelmingly.** Where gravity is strong compared to a_0 , the correction dies off *exponentially*. At Earth’s orbit around the Sun, the predicted deviation carries a suppression factor of about 10^{-3457} — a decimal point followed by three and a half thousand zeros. The theory does not “approximately” recover the solar system;

it recovers it with astronomical overkill. Planetary position tables (ephemerides), spacecraft tracking, lunar laser ranging: untouched.

- **Below the threshold, the boost** — precisely the behavior that makes rotation curves flat and gives galaxies the observed tight relation between visible mass and rotation speed (the baryonic Tully–Fisher relation), which in this theory is a theorem, not a fit.
- **In between, one smooth curve** — and this is where the theory faces its most granular test.

That test is the radial acceleration relation of Section 2: thousands of measured points from the SPARC sample of 175 galaxies, each point a direct comparison of observed gravity against visible-matter gravity. Fit with the framework’s own interpolation and its dark-energy-anchored a_0 , the scatter of the data around the predicted curve is **0.108 dex** — dex is the astronomers’ unit of “factors of ten,” and 0.108 dex means the typical point sits within about 28% of the curve, over data spanning factors of thousands. For calibration: the same data fit with the conventional MOND curve used in the literature gives 0.122 dex. **The dark-energy-anchored version fits slightly better than the standard one.** It is worth pausing on that: anchoring a_0 to an independently measured cosmological quantity *removes* a free parameter, and the fit does not degrade — it improves.

One more galaxy-scale number, because later sections lean on it: the theory’s third term (the cluster mechanism of Section 10) also technically operates inside galaxies, so one must check it does not spoil this agreement. It contributes 0.034% of a typical galaxy’s visible mass in the interior — shifting the radial acceleration relation by 0.0004 dex, roughly 270 times smaller than the observed scatter. Galaxies do not notice it.

7. Light: the test that kills theories like this

Gravitational lensing deserves its own short section, because it is the single most efficient executioner of modified-gravity theories — it personally killed the previous arm of this program — and because the way this theory passes is clean enough to state exactly.

When light from a distant galaxy passes a massive object, its path bends; astronomers measure the mass of the deflector from the bending. Separately, they can measure the same object’s mass from the orbits of things moving around or within it. In general relativity plus dark matter, both measurements see the same total mass, so they agree — and observationally, **they do agree**, across galaxies and clusters. Any theory in which the “orbit mass” and the “lensing mass” differ is dead.

In this theory they agree *identically*, for a structural reason. The khronon and aether do their work by shaping the same spacetime geometry that matter and light both respond to; light couples to that geometry exactly as in Einstein’s theory, with nothing extra. In the technical language, the two gravitational potentials that govern slow matter and light are forced equal — the parametrized-post-Newtonian parameter γ equals 1 exactly — so whatever total gravity the theory produces, orbits and lensing see the same thing, automatically. The 21-sigma execution that ended the modified-inertia arm becomes, in this arm, a 0.6-sigma pass — statistical agreement.

The physical picture is worth one sentence: in this theory the dark field genuinely *gravitates* — it is a real source of spacetime curvature where it accumulates — rather than being a bookkeeping correction to matter’s response, and that is precisely why light cannot tell the difference between this universe and a dark-matter one at the lensing level.

8. Job two: the universe — dark energy and “dust” from one function

The second piece of the function, $K(Q)$, depends only on how fast the khronon ticks. It has to manage the entire cosmological side of the ledger, and it does three jobs with three features of one curve.

Feature one: the floor. The function K has a constant offset: a fixed energy floor of exactly the dark-energy density. A constant energy density filling space, unchanging as the universe expands, pushing outward with a pressure exactly equal and opposite to its energy density (the relation cosmologists write as $w = -1$): that *is* dark energy, by definition. In this theory it is not added as a separate ingredient; it is the resting value of the khronon’s own energy function — the value the field’s energy takes when it sits at its natural minimum. At that minimum, $w = -1$ holds *exactly*, proved, not tuned.

Feature two: the wiggle. Displace the khronon’s ticking rate slightly off its natural value and the energy rises — and, crucially, this excitation energy *dilutes as the universe expands exactly the way ordinary matter does*, with zero pressure. Cosmologists call anything with that behavior “dust.” The universe’s dark matter, as far as cosmological measurements are concerned, is dust — and here, the dust is not a substance added to the theory: it is the *same field*, slightly excited. How much excitation the universe carries is set by initial conditions (one number, playing the role Λ CDM’s dark-matter density plays), and Section 9 is honest about what that does and does not buy.

Feature three: the cap — and the theorem. The obvious, simplest choice for K — a plain parabola, energy rising as the square of the displacement — was already studied by others, including by Skordis’s own collaborators, and in 2024 Blanchet and Skordis published what amounted to a no-go: with the parabola, matching cosmology and matching galaxies impose requirements on one of the theory’s mass scales that conflict by a factor of 455. The natural conclusion was that this class of theories cannot do both jobs. This program’s response was first to prove a small theorem: for *any* power-law K — quadratic, quartic, sextic, anything — the early universe comes out wrong in a specific calculable way. As you run the universe backward and the khronon’s displacement grows, a power-law energy function makes the field’s pressure blow up: the early cosmos fills with a stiff, honey-like medium that is flatly ruled out by the observed abundances of the lightest elements (hydrogen, helium) forged in the universe’s first minutes, and by the microwave background. The exponent only changes how badly. **No power law can work. What works is boundedness:** give the energy function a hard cap — a maximum displacement it can never exceed, approached with a square-root shape exactly like the one special relativity uses to cap velocities at the speed of light (the technical name is a DBI form, borrowed from string theory’s toolbox). With the cap in place, run the universe backward and the pressure stays finite; the early-time behavior turns back into harmless dust. And the $455\times$ conflict does not merely shrink: the two requirements stop conflicting and start pointing the same way, with a 226-fold margin to spare. The published no-go, under this program’s choice of function, dissolves.

The receipt. Words like these are cheap; the test is numerical. The theory was run through CLASS — one of the two standard precision codes every working cosmologist uses to compute the microwave background — with the capped K in place. Result: the acoustic peak pattern of the CMB agrees with the standard model’s to **0.069%**, far inside measurement error, and the matter distribution at the representative measured scale to 1.7%. On the universe’s oldest light, this

theory and Λ CDM are indistinguishable. Two disclosed caveats accompany that pass, and Section 15 states them (they concern the *later* growth history, verified only in part, and a parameter window tightened by the health checks of Section 11 — tightened, notably, by this program’s own hand against its own earlier choice).

9. So is there dark matter in this theory or not?

This question deserves a section with no hedging, because it is the question, and because the answer’s honesty is the program’s credibility.

Start with what “dark matter exists” actually asserts: that the universe’s inventory contains an undiscovered *substance* — a new species of particle, sitting in halos, waiting in the dark. That is the picture the words smuggle in, and the picture direct-detection experiments have spent forty years failing to confirm.

In this theory that picture is replaced, not merely relabeled. Λ CDM adds two unrelated dark ingredients to the universe: a constant (dark energy) and a material (cold dark matter, a particle species). This theory adds no new species of matter. It adds one dynamical field — the khronon, with its aether companion — and that single field’s three behaviors *are* the three dark phenomena: its resting energy is dark energy, its gentle excitation does the cosmological job dark matter was invented for, and its spatial response is the MOND law.

Now the exact statements, each with its precise status:

There is no dark matter particle. Nothing in this theory is a new species of matter. No WIMP, no axion, nothing for an underground detector to catch or a collider to make. Four decades of null results are, in this theory, exactly what should have been found: there is nothing of that kind to find.

The CMB never detected a substance in the first place. A theorem in this program’s corpus makes this precise: the microwave background constrains a *behavior*, not an identity. It demands something that gravitates, exerts no pressure, and dilutes like matter — and it cannot distinguish, even in principle, whether that something is a particle fluid or a field excitation: the two differ by exactly zero sigma in the CMB’s eyes. “Dark matter,” in cosmology, is a job description, not an identified employee. Λ CDM fills the job with a hypothetical particle. This theory fills it with the field that is already on the payroll for dark energy and MOND. The job itself is real and non-negotiable: delete the pressureless component and the ratio of the CMB’s third to first acoustic peak shifts by 54%, a catastrophe no refitting of other parameters can absorb. So the flat sentence “there is no dark matter” is not available to this theory or to any theory that matches the CMB, and this document will not use it.

And in galaxies — the honest status is: open, and sharply posed. The galaxy data themselves are unambiguous: rotation curves are fit by visible matter plus the modified law alone, to 0.108 dex, with no dark contribution needed or wanted. For the theory to be right, then, its own cosmic dust — the field excitation of Section 8, which certainly exists at cosmological scales — must largely *stay out* of the inner regions of galaxies where those curves are measured. Does it? This program had a mechanism that guaranteed it, and killed that mechanism itself this week (Section 10 tells the story). What remains are two computable branches: in one, the settled field inside a galaxy’s halo relaxes to a flat, centrally-evacuated profile — a configuration this theory’s

own equations prefer in equilibrium — and galaxies are safe by orders of magnitude; in the other, the dust piles up the way cold particles would, and the theory *overshoots* its own best data. Which branch nature’s version of this theory takes is a hard nonlinear calculation — the kind nobody, in any group, has yet performed for this class of theory — and it is now the sharpest open problem on the program’s list, stated in Section 15 and scheduled as the next calculation.

So the defensible summary — the one this program will defend in any referee report — is:

Dark matter is not a thing in this theory: no particle, no new substance — one field doing three jobs. The cosmological job that Λ CDM assigns to a particle is done here by the same field that supplies dark energy and the galaxy law. Whether that field’s dust stays out of galaxy interiors, as the data demand, is the theory’s sharpest open problem — with both branches computable and the calculation scheduled.

Whether nature agrees is Section 13’s business.

10. Job three: clusters — the thorn, the failures, and the live candidate

Galaxy clusters — swarms of hundreds of galaxies in halos of hot gas, the largest bound objects in existence — have been MOND’s thorn for forty years, and this program spent its hardest weeks on them. This section tells the story as it happened, including the dead bodies, because the dead bodies are why the surviving candidate deserves attention.

The problem. Even with MOND’s boost, clusters come up short: the gravity needed to hold the hot gas and the galaxies exceeds what the visible matter plus the modified law can produce, by around a factor of two. Milgrom’s rule, which clears every galaxy, hits a wall at the next scale up.

The failures, in the open. This program tried, and killed, three mechanisms in sequence — each kill executed by the program’s own scripts, each published, each now on a formal do-not-cite list:

1. *A geometric lever* based on how the field’s response scales with the size of the system. Killed: its required mass scale conflicts with independent galaxy-lensing bounds published by Misteale, McGaugh and Hossfelder — it satisfies neither of the two bounds it needed to.
2. *A primordial route*: clusters sit where the early universe was clumpiest — perhaps the dark field remembers that. Killed three independent ways, the third decisive: a smoothness theorem. Any suppression mechanism of the proposed kind removes *fluctuations*, not the *mean*, and a collapsing region captures all the cold material in its basin regardless — so the proposed difference between clusters and galaxies provably cannot arise from initial conditions of that class. The route’s own published write-up was amended to say so. (This kill is also what reopened the galaxy question of Section 9: the same theorem says the field’s dust falls into *every* collapsing basin, galaxies included, unless the settled field arranges itself away from their centers — the open calculation described there.)
3. *A potential-depth response*: let the field respond to how deep a gravitational well it sits in — clusters are the deepest wells, so target them by depth. Killed by a scale-freeness check the program’s own specification had missed: gravitational potentials in the cosmos are nearly the same depth everywhere on large scales (about one part in a hundred thousand), so a response steep enough to single out clusters floods the whole universe with twenty times too much energy. Written down, checked, dead within a day.

The survivor. The third failure taught the lesson that built the candidate. Depth alone cannot select clusters; scale alone cannot either (the second failure); but there is one property clusters uniquely combine. Recall a_0 , the threshold acceleration. Galaxy interiors sit *above* it; the cosmic web at large sits *far below* it; but the bodies of galaxy clusters — at the radius where their masses are actually measured — sit at 0.33 to 0.58 times a_0 . **Clusters are the cosmic objects that live at the MOND transition itself.** They are the unique environments that are simultaneously deep and *at the threshold*.

So the third piece of the function is a *resonance at the transition*. It couples the khronon’s ticking to its spatial variation through a bell-shaped profile B that peaks exactly where local gravity equals a_0 and dies off on both sides. Like a wine glass that answers only to its own note, the dark sector’s extra response answers only to environments near a_0 — which is to say, clusters. The elegance one should insist on: **the location of the peak costs nothing.** The theory’s Y -variable is already naturally measured in units of a_0^2 — the peak position is inherited from the framework’s central number, not tuned. What is genuinely new and fitted is one amplitude, called A , calibrated so clusters get their missing gravity.

Any such mechanism must then survive everywhere else, and this one was marched through five environments: **clusters** (calibration — that one is free); **galaxy interiors** (contributes 0.034% of the visible mass; the radial acceleration relation moves by 0.0004 dex — unnoticeable); **galaxy outskirts** at about three million light-years (1.2% extra effective mass, versus a measured lensing tolerance of 5.9% — inside bounds by a factor of about five); **the smooth universe** (0.6% of the mean matter density — harmless, and the term vanishes identically in exact cosmological symmetry, so the CMB results of Section 8 are untouched *by construction*); **the solar system** (a correction of order 10^{-19} — nineteen zeros — of nothing).

And it evades both earlier executioners by design: it is a *response*, not a transported substance, so the smoothness theorem does not apply (nothing is being carried in from the early universe); and it couples to the environment directly, so it needs no primordial selection.

Status, stated coldly: a candidate, not a result. It has passed the environment matrix above, a dedicated stability audit, and — as of this document’s date — the full perturbation analysis of Section 11. But it has not been fit to real cluster mass profiles at all — by this program or anyone; the amplitude is calibrated on a single summary number, and a real cluster model is owed. Its amplitude also sits in tension with a separate published cluster analysis by Misteale and collaborators (not the lensing bounds of failure 1) at that analysis’s demanding end — Section 11 turns exactly this into a kill test. And this program has been wrong about clusters three times this month. The reader should hold it accordingly. But it is the first cluster mechanism this program has produced that *survived its own execution squad* — and its slogan states a genuinely new idea, whatever its fate: *the dark sector’s response is resonant at the MOND transition, and clusters are the objects that live at a_0 .*

11. Could the whole thing be sick? The safety inspection

A field theory can fit every observation and still be fatally diseased. This section explains the diseases, in plain terms, and reports the inspection — because “we checked whether our own theory is sick, using tests that could have killed it” is precisely what separates a theory from a wish. All of this is recent work: an isolated-term health check first, then the full analysis with every interaction

included, completed the day of this document.

Ghosts. The gravest disease. A “ghost” is a field configuration carrying *negative* kinetic energy. If a theory has one, the vacuum itself is unstable: empty space can spontaneously erupt into pairs of positive- and negative-energy excitations, endlessly, everywhere — the theory predicts that the universe should instantly self-destruct, which is a strong prediction and a wrong one. Historically, ghost diseases have killed more modified-gravity proposals than any observation has. The base AeST framework was proven healthy by its authors. The question was whether this program’s additions — above all the new cluster term — poison it. The answer is a theorem: the cluster term’s contribution to the theory’s kinetic structure is *provably incapable of subtraction* — it can add stability but never remove it, in any environment, for any field values. Healthy plus non-negative is healthy. **The bump cannot create a ghost — ever.** Proved symbolically in closed form, then verified numerically on two hundred randomized configurations.

The speed of gravitational waves. In 2017, two neutron stars collided 130 million light-years away. The ripple in spacetime and the flash of light arrived within two seconds of each other — after 130 million years in transit. Gravitational waves therefore travel at the speed of light to about one part in a quadrillion, and that single measurement annihilated whole families of modified-gravity theories the week it was announced. In this theory, the new terms touch the relevant part of the geometry only algebraically — not through the derivatives that set a wave’s speed — and the wave speed works out to exactly the speed of light: not approximately, *identically*, as a structural fact. **This theory could not have failed the 2017 test even in principle.**

Gradient stability, and a bound with teeth. Beyond ghosts there are subtler instabilities, where small spatial ripples of the field amplify instead of oscillating. Here the inspection *bit*, twice — and the bites are the best evidence of its sincerity. First: on the smooth cosmological background, health of the cluster term demands that the cap parameter of Section 8 be small — quantitatively, at most about 10^{-6} in the theory’s own units — which *overrules a value this program itself had previously used* (10^{-2}). The theory’s own consistency disciplined its author; the allowed window remains 3.7 orders of magnitude wide, and nothing previously verified is disturbed, but the old fiducial number is dead, by the program’s own hand. Second: inside cluster-like environments, the stability analysis caps the cluster term’s amplitude A from above. With every field interaction included, the cap lands between roughly 2.7 and 7.4 times the calibrated value (the residual range awaits one coefficient from Skordis and Złośnik’s published supplemental material — a literature lookup, not a new calculation). Meanwhile the separate cluster analysis by Misteale and collaborators mentioned in Section 10 would demand, at its far end, an amplitude 34 times the calibrated value. **Stability forbids that end outright** — it exceeds the cap several times over — while the near end of the same analysis, 4 times calibrated, sits at the boundary: alive or dead by a coefficient. The theory is thus *pinched from two sides*: observation demands the amplitude be big enough, internal health forbids it from being too big, and the window between is narrow and closing. A theory that arranges its own execution window is doing falsifiability properly.

Loose ends, disclosed. Two flags remain open and are stated wherever the results are: a possible mild instability in the aether (vector) sector deep inside galaxies — suppressed by a small coupling, flagged for a dedicated analysis rather than resolved; and one known curiosity of the base AeST framework (a formal unboundedness below a certain wavelength, known to its authors) which, at this program’s parameter values, migrates to the scale of the entire observable universe — an amusing and possibly meaningful relocation, noted, not celebrated.

The inspection’s summary: **no ghosts, by theorem; gravitational waves exactly at light**

speed, by construction; stability bounds that bite the theory’s own parameters, accepted and published; two flags open and labeled.

12. What it costs: the price tag, read aloud

No honest sales pitch omits the invoice.

Parameters. Λ CDM’s dark sector runs on two numbers: how much dark energy, how much dark matter. This theory runs on five: the vacuum scale (dark energy’s density), the amount of khronon excitation (the dust’s abundance — an initial condition, not explained), the field’s mass scale, the cap scale of Section 8, and the cluster amplitude of Section 10. Five against two. **This is not a simpler theory, and this document will not pretend otherwise.** What is bought for the extra three: one function replaces three unrelated sectors; the galaxy-scale acceleration a_0 is not an *additional* dial on top of those five — it is re-expressed through the field’s own condensate mass, a re-expression that checks out arithmetically to 0.076% but is a relabelling, not an independent derivation (Section 15, item 1); and the theory acquires kill-tests that Λ CDM, with its flexibility, does not offer.

A preferred frame. The aether restores a dynamical version of the notion of absolute rest that special relativity abolished. The observational cost is confined by the same structure that saves lensing, but precision solar-system and pulsar bounds on preferred-frame effects exist, and Section 15 lists what has not yet been computed against them.

An inherited tension. The Cassini spacecraft’s radio-link measurements constrain one post-Newtonian quantity at a level in 3-to-15-sigma tension with this class of kernels. Inherited, disclosed, unresolved. It sits on the worry list, neither hidden nor fatal at current precision of the theory-side calculation.

Things given up. Earlier arms of this program claimed more: a derivation of a_0 ’s coefficient from first principles (failed — κ is measured); a laboratory-testable modified-inertia effect (that arm is dead); a theory-of-everything ambition (publicly retracted in June 2026, and the retraction stands — Section 15). The surviving theory is narrower than the program’s earliest hopes, and better for it.

13. How it dies: the predictions

A theory is only as good as the experiments that could kill it. This one maintains a registered kill-list. The word “registered” is doing real work in that sentence: the galaxy-survey predictions below were *frozen before the data arrive* — written into a public preregistration document whose cryptographic hash (a digital fingerprint that changes if a single character is edited) is committed publicly, so the targets cannot be quietly moved after the fact. Nine amendments to that document exist; each is itself hash-stamped; the analysis pipeline is committed alongside. When the data land, the confrontation is automatic and the escape hatches are welded shut.

One: wide binary stars (the sharpest, and it is imminent). Pairs of stars orbiting each other at great separations — thousands of times the Earth–Sun distance — have mutual gravity right at a_0 . They are the cleanest MOND laboratory in the sky, free of the messy gas dynamics of galaxies. The European Gaia satellite’s fourth data release will provide orbital statistics for enormous numbers

of them. This theory’s registered prediction: orbital velocities boosted over Newton by a factor of 1.2139. (A stated alternative, 1.2592, corresponds to the other of two defensible conventions for anchoring the dark-energy density — both were frozen in the registration, so a result matching neither kills the theory regardless of convention.) The dead modified-inertia arm predicted 1.1582 — separated from the live target at about 2.7 sigma by the survey’s expected precision: enough for DR4 to distinguish the arms. The numbers are frozen in public, by design.

Two: a directional signature (already flickering). Because the aether picks out a frame and the external galactic field matters in this theory (the “external field effect,” a MOND signature with no Λ CDM counterpart), binaries oriented differently with respect to the galaxy should behave measurably differently, at the 1-to-4% level, with a definite sign. The dead arm predicted exactly zero such effect. A first statistical test on existing data gives a positive signal with the predicted sign at $p = 0.029$ — suggestive, not decisive (about one-in-thirty odds of arising by chance, and one such firing is not a discovery). It is listed because it was predicted *before* it was seen, and DR4 will settle it.

Three: clusters, from three sides (the candidate’s own gallows). The Section 10 mechanism must produce anisotropic stress in clusters — a specific mismatch, of order tens of percent, between cluster masses inferred from lensing and from hot-gas pressure profiles. Both measurement types exist and are improving; a dedicated confrontation is a data-analysis project, not a new theory. The mechanism also predicts *cluster-to-cluster variety* in the dark residual — different clusters sit at different distances from the a_0 resonance — where mechanisms tied to a fixed geometric scale predict uniformity; existing cluster samples can already test this. And the amplitude is pinched between an observational floor and a stability ceiling (Section 11), with the gap at the level of one pending coefficient. Any of the three jaws can close.

Four: the redshift evolution of a_0 . If a_0 is truly anchored to dark energy, it must track the dark-energy density through cosmic time — a specific, gentle evolution law, published by this program, distinct both from a constant a_0 and from naive expansion-rate scaling. High-redshift rotation measurements are near the required precision now, and current claims in the literature cut both ways; the program’s assessment of its own standing here is “contested, being watched,” and the prediction is on record.

Five: two smaller flags on the record. The aether sector implies a specific, computable pattern of tiny Lorentz-violation effects in precision laboratory experiments (the g-2 family of tests) — registered as a prediction target. And a modified-gravity universe forms massive structures *earlier* than Λ CDM expects — a long-standing MOND expectation that the James Webb Space Telescope’s surprisingly mature early galaxies currently favor, noted with the caution that early JWST results are still being digested by everyone.

Six: no particle, ever. Every future direct-detection experiment continues to find nothing. An unfalsifiable-sounding prediction with real content: a single confirmed dark-matter particle detection kills this theory outright. Forty years of null results are, so far, forty years of this prediction coming true.

14. Why should anyone trust this? The receipts

This program is one person with computers, working outside the academy, in a field with a long tradition of confident outsiders being wrong. That context earns extra skepticism, and the program’s

answer to it is procedural, not rhetorical. Four procedures:

Every claim has a script. Every load-bearing number in this document — every sigma, every dex, every bound — is computed by a named script in the public repository, runnable by anyone, that re-derives the number from scratch and *exits with an error* if any of its internal checks fail. The technical paper’s results table names the script for every row; scripts carry negative controls (deliberate wrong inputs that must fail, proving the test can fail). The repository currently runs to hundreds of such checks, green.

Predictions are frozen before data. The preregistration-and-hash discipline of Section 13: the sharpest predictions are cryptographically time-stamped in public before the deciding dataset exists. Whatever DR4 says, the record of what was predicted cannot be edited afterward.

Errors are published louder than results. The program maintains a formal do-not-cite list of its own withdrawn claims — currently more than a dozen entries, including the three dead cluster mechanisms of Section 10, a wrongly-claimed statistical exclusion (an error in the program’s own kernel arithmetic, found by the program, announced by the program), and a retracted theory-of-everything overclaim (Section 15). Several corrections ran *against* the program’s interest; at least one ran in its favor and was published with exactly the same prominence. The week’s work log recorded in the technical corpus includes roughly a dozen near-misses — moments where an exciting wrong number nearly got reported — caught by the standing rule that every important quantity be computed a second, independent way.

The competition is credited by name. The base framework is Skordis and Złóśnik’s. The transition function is Milgrom’s. The data are the SPARC team’s. The sharpest published objections — the ones this program spent weeks answering — are Blanchet and Skordis’s no-go and the Mistele–McGaugh–Hossenfelder lensing bounds, and both are cited as the serious challenges they are. The claim defended as original is narrow and stated in the technical paper’s final section: the dark-energy anchor and its coefficient, the choice and factorization of the function $\mathcal{F}$, the bound-ness theorem and the dissolution of the $455\times$ no-go, the cluster resonance mechanism, and the health-and-matrix theorems of Section 11.

None of this makes the theory *right*. It makes the theory *checkable*, which is the only coin science accepts.

15. What is NOT claimed

The technical paper’s most important section is its list of non-claims. Reproduced here in plain words, it is the fence past which no honest summary of this work may step:

1. $\kappa = \frac{1}{2}$ **is not derived.** The coefficient linking a_0 to dark energy is *measured*: 0.551 ± 0.043 (and ± 0.063 if the current tension in measurements of the universe’s expansion rate is carried as an additional systematic — the coefficient is hostage to that dispute, since it scales one-for-one with it). One-half sits comfortably inside the error bar; so do other simple candidates, including $1/\sqrt{3}$. Every attempted first-principles derivation either failed or collapsed into circularity, and the technical corpus proves several *cannot* work. One well-posed open problem remains (Section 16).
2. **“Zero dark matter” is not claimed.** Section 9 is the claim: no particle, no new substance, one field — but the cosmological pressureless job is real and filled at full strength. 2b. **“No**

dark matter in galaxies” is currently an open problem, not a result. The mechanism this program had for keeping the field’s dust out of galaxy interiors was withdrawn this week, by the program’s own theorem. The favorable branch (the field’s settled configuration is centrally evacuated) and the fatal branch (it piles up like cold particles and overshoots the galaxy data) are both computable; the nonlinear calculation that decides between them has never been done for this class of theory, by anyone, and is the program’s next scheduled work.

3. **The cluster mechanism is a candidate, not a result.** It has survived this program’s own execution attempts — environment matrix, health check, full perturbation analysis — and no external scrutiny. It has not been fit to real cluster mass profiles by anyone, and its amplitude tension with a published cluster analysis is live and could kill it.
4. **The late-universe growth history is only partly verified.** The CMB acoustic-peak pass is a real CLASS-code result; the subsequent growth of cosmic structure needs a dedicated evolving- sound-speed treatment that is owed, not done. One initial-condition confrontation (with intergalactic-hydrogen absorption data) is likewise owed.
5. **Solar-system preferred-frame parameters are not yet computed** for this specific function choice, against lunar-ranging and pulsar bounds; and the Cassini tension of Section 12 is inherited, disclosed, and unresolved.
6. **The wide-binary prediction is the idealized asymptote.** The registered 1.2139 is the clean point-source limit; the full nonlinear treatment is owed before DR4 lands.
7. **This is not a theory of everything.** It contains no explanation of particle physics, no quantum gravity, no unification. An earlier overclaim in that direction was publicly retracted on 2026-06-23, and the retraction stands. This is a proposed field theory of the dark sector, anchored to one measured coincidence — nothing more, and the program regards “nothing more” as already ambitious enough.

16. The one open problem of the coefficient

Strip away everything settled and one question about the *number* remains, and it can now be stated in a single line — which is itself progress, because for most of this program’s life the question was a fog.

A calculation in the technical corpus treats a body accelerating through the bath of gravitational-wave fluctuations that a dark-energy-dominated universe must contain — must, because the horizon of Section 3 carries a temperature, and a temperature means the universe is filled with a faint thermal jitter, including a jitter of spacetime itself. Two remarkable things happen in the calculation. First, an apparently hopeless suppression — effects of this kind are normally crushed by the Planck scale, quantum gravity’s remoteness, to the tune of hundreds of orders of magnitude — is *exactly cancelled* by the enormous entropy of the cosmic horizon: the product of the horizon’s entropy and its curvature is π , identically, not approximately. What survives the cancellation is a formula of exactly the right form: a_0 equals c times the horizon expansion rate times a pure number. Second, the pure number — and with it κ — reduces to one well-posed question: $\kappa^2 = 8\pi \times \varepsilon_{\text{tot}}$, **where ε_{tot} is the fraction of the horizon’s fluctuation modes that act on a slowly accelerating body.** If ε_{tot} could be computed and came out to $1/(32\pi)$, then $\kappa = \frac{1}{2}$ exactly, and the last measured number in the theory would become a derived one.

It has not been computed. Five defensible ways of setting up the count give answers spanning a factor of about 160, one of them lands exactly on $\frac{1}{2}$, and that one was checked and found to rest

on an invalid step — a near-miss the program documented rather than published. The question is stated, sharp, and open. It is either this framework’s Rosetta stone or its tombstone inscription, and the program genuinely does not know which.

17. Where this stands

As of 2026-08-09, the theory described here is, to this program’s knowledge, the only proposal in the literature that simultaneously: anchors MOND’s acceleration scale to the measured dark-energy density; embeds it in a relativistic field theory that passes gravitational lensing with exact equality of dynamical and lensing mass; reproduces the CMB acoustic peaks in a real Boltzmann-code run; recovers the solar system with exponential overkill; fits the radial acceleration relation slightly better than standard MOND with one fewer free scale; carries a stability-vetted candidate for clusters resonant at its own threshold; and holds a certificate that its additions cannot introduce ghosts, with gravitational waves at exactly light speed.

It is five parameters against Λ CDM’s two. It is one outsider’s program against a standard model with tens of thousands of person-years behind it. It has been wrong, in public, more than a dozen times this year, and says so — and its two sharpest open problems (the dust-in-galaxies calculation of Section 9; the mode-count of Section 16) are stated in print, with the failure branches spelled out. And it is now, in the only sense that matters, *finished as a construction*: the action is written, the safety certificate is signed, and everything that remains is calculation and confrontation — confrontations it has pre-committed to losing publicly if it loses.

The universe’s oldest light, its strangest galaxies, and its emptiest places may yet converge on the same verdict they have always given Newton and Einstein: *close, but the truth was stranger*. This theory’s wager is narrow and specific: that the strangeness begins at one hundred-billionth of a g — at the acceleration where a falling body starts to feel the edge of the universe — and that the number written on that edge is, within a hair’s measurement, one-half.

18. Credit

The intellectual lineage, so no reader mistakes what is borrowed for what is claimed: MOND is Milgrom’s (1983), and the transition function this program derived from de Sitter space turns out to be exactly Milgrom’s own 1999 formula. The classical field-theoretic MOND is Bekenstein and Milgrom’s (1984). The relativistic framework AeST is Skordis and Złośnik’s (2021), including its CMB success; the ghost-condensate technology traces to Arkani-Hamed, Cheng, Luty and Mukohyama (2004). The galaxy data are the SPARC compilation of Lelli, McGaugh and Schombert. The published challenges answered here are Blanchet and Skordis’s (2024) and Misteale, McGaugh and Hossenfelder’s (2023). The DBI functional form is borrowed from string theory’s standard toolkit.

Original to this program: the coefficient relation $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ as an anchor with κ measured at 0.551 ± 0.043 ; the factorized function $\mathcal{F}$ and its offset-DBI Q-sector; the boundedness theorem and the dissolution of the $455\times$ no-go; the CLASS confrontation of that sector; the a_0 -bump cluster mechanism and its five-environment, health, and full-matrix results; the preregistered wide-binary program with its hash-stamped amendments; and the open problems of Sections 9 and 16.

Glossary

- **a_0** — the threshold acceleration, 9.36×10^{-11} m/s², below which gravity departs from Newton. About one hundred-billionth of Earth’s surface gravity.
- **Acoustic peaks** — the characteristic pattern of spot sizes in the cosmic microwave background, left by sound waves in the primordial plasma; measured exquisitely, and the sharpest test of any cosmological theory.
- **Action** — the single master formula defining a physical theory; all equations of motion follow from it by a standard procedure.
- **AeST** — Aether-Scalar-Tensor theory (Skordis & Złośnik 2021), the relativistic framework this theory lives in.
- **Aether** — here, a field of unit arrows pointing into the future at every point of spacetime, defining a local standard of rest; dynamical, not rigid.
- **CLASS** — one of the two standard precision codes for computing cosmic-microwave-background predictions; used here to test this theory, not a simulation of it.
- **Dark energy** — the smooth ingredient accelerating the universe’s expansion; about 68% of the cosmic energy budget; in this theory, the resting energy of the khronon field.
- **DBI form** — an energy function with a hard cap approached via a square root, mathematically parallel to how special relativity caps speeds at c .
- **dex** — astronomers’ unit for factors of ten; $0.1 \text{ dex} \approx 26\%$.
- **Ghost** — a field excitation with negative kinetic energy; fatal, because it makes empty space unstable. This theory carries a proof it has none.
- **Kernel** — this program’s word for the chosen transition curve between the Newtonian and modified regimes.
- **Khronon** — the scalar field at the theory’s heart, behaving like a universal clock-reading; its states supply dark energy, the cosmic dust, and the MOND effect.
- **Λ CDM** — the standard model of cosmology: general relativity plus dark energy (Λ) plus cold dark matter (CDM).
- **Lensing** — the bending of light by mass; measures gravity with light instead of orbits.
- **MOND** — Modified Newtonian Dynamics (Milgrom 1983): the hypothesis that the galaxy anomaly reflects a change in the law below a_0 rather than unseen matter.
- **Parsec** — astronomers’ distance unit, about 3.26 light-years.
- **Preregistration** — publicly freezing a prediction, with a cryptographic fingerprint, before the deciding data exist.
- **Radial acceleration relation (RAR)** — the observed tight universal curve linking total gravity to visible-matter gravity across all galaxies; this theory fits it to 0.108 dex.
- **Sigma (σ)** — a measure of statistical confidence; 5σ (about one-in-3.5-million odds of chance) is physics’ conventional discovery threshold, and 21σ — the lensing kill of this program’s earlier arm — is beyond-argument territory.
- **SPARC** — the reference sample of 175 galaxies with precision rotation curves and photometry, compiled by Lelli, McGaugh and Schombert.
- **Wide binaries** — star pairs separated so widely their mutual gravity sits near a_0 ; the cleanest test environment for MOND-type laws, and the subject of this program’s registered Gaia DR4 prediction.